MTH309 PRACTICE PROBLEMS WEEK 1

Question 1. Given a function $X : \Omega \to \mathbb{R}$ on a non-empty set Ω, recall the definition of the pre-images $X^{-1}(A), A \subseteq \mathbb{R}$. Prove the following properties:

(i) $X^{-1}(\mathbb{R}) = \Omega$.

(ii) $X^{-1}(\emptyset_{\mathbb{R}}) = \emptyset_{\Omega}$, where $\emptyset_{\mathbb{R}}$ and $\emptyset_{\Omega}$ denote the empty sets in $\mathbb{R}$ and Ω, respectively. (When there is no chance of confusion, we simply write $X^{-1}(\emptyset) = \emptyset$.)

(iii) For any two subsets A, B of $\mathbb{R}$ with $A \cap B = \emptyset$, we have $X^{-1}(A) \cap X^{-1}(B) = \emptyset$.

(iv) For any subset A of $\mathbb{R}$, we have $X^{-1}(A^c) = \left(X^{-1}(A)\right)^c$.

(v) Let $\mathcal{I}$ be an indexing set. For any collection $\{A_i : i \in \mathcal{I}\}$ of subsets of $\mathbb{R}$, we have

$$X^{-1}\left(\bigcup_{i\in\mathcal{I}} A_i\right) = \bigcup_{i\in\mathcal{I}} X^{-1}(A_i), \quad X^{-1}\left(\bigcap_{i\in\mathcal{I}} A_i\right) = \bigcap_{i\in\mathcal{I}} X^{-1}(A_i).$$

Question 2. Suppose that $\mathcal{F}$ is a collection of subsets of Ω with the following properties: $\Omega \in \mathcal{F}$ and $A, B \in \mathcal{F}$ implies $A \setminus B := A \cap B^c \in \mathcal{F}$. Show that $\mathcal{F}$ is a field.

Question 3. Suppose that a collection $\mathcal{F}$ of subsets of Ω has the following properties.

(1) $\emptyset \in \mathcal{F}$.

(2) If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.

(3) If $A \in \mathcal{F}$, then A^c is a finite disjoint union of members of $\mathcal{F}$.

Consider another collection $\mathcal{C}$ consisting of finite disjoint union of members of $\mathcal{F}$. Show that $\mathcal{C}$ is a field of subsets of Ω.

Question 4. Let $\mathcal{F}_\alpha, \alpha \in \Lambda$ be an arbitrary collection of σ-fields on a non-empty set Ω. Show that $\mathcal{F} := \bigcap_{\alpha\in\Lambda} \mathcal{F}_\alpha$ is also a σ-field.

Question 5. Consider the following collection of subsets of $\mathbb{R}$.

$$\mathcal{F}_1 := \{A \subseteq \mathbb{R} : \text{either } A \text{ is countable or } A^c \text{ is countable}\}.$$

(i) Verify that $\mathcal{F}_1$ is a σ-field.

(ii) Consider the following two σ-fields.

$$\mathcal{F}_2 := \sigma\left(\{x\} : x \in \mathbb{R}\right)$$

and

$$\mathcal{F}_3 := \sigma\left(A : A \text{ is a countable subset of } \mathbb{R}\right).$$

What is the relation between $\mathcal{F}_1, \mathcal{F}_2$ and $\mathcal{F}_3$?

Question 6. Let $A_1, A_2, \cdots, A_n$ be subsets of a non-empty set Ω. Show that the σ-field generated by $A_1, A_2, \cdots, A_n$ has at most 2^{2^n} sets.

Question 7. Let $f : \Omega \to \Omega'$ be a function and let $\mathcal{F}$ be a σ-field in Ω'. Show that $f^{-1}(\mathcal{F}) = \{f^{-1}(A) : A \in \mathcal{F}\}$ is a σ-field in Ω.

Question 8. Show that open sets in $\mathbb{R}$ can be written as a disjoint union of finitely or countably many open intervals.

Question 9. Show that the Borel σ-field $\mathbb{B}_{\mathbb{R}}$ is countably generated, i.e. there exists a countable collection $\mathcal{C}$ of subsets of $\mathbb{R}$, such that $\sigma(\mathcal{C}) = \mathbb{B}_{\mathbb{R}}$.